



OR – Technological Project

Final Report

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1 Mixed-Integer Quadratic Programming model

$$\max_{w, z, \Delta} \quad \mu^T w - \gamma w^T \Sigma w$$

s.t.

$$\text{Fully Invested:} \quad \sum_{i=1}^n w_i = 1$$

$$\text{No Short-selling:} \quad w_i \geq 0, \quad \forall i \in 1, \dots, n$$

$$\text{Maximum Individual Weight:} \quad w_i \leq 0.10, \quad \forall i \in 1, \dots, n$$

$$\text{Maximum Portfolio Beta:} \quad \sum_{i=1}^n \beta_i w_i \leq 1.0$$

$$\text{Minimum Individual Weight :} \quad w_i \geq 0.005 \times z_i$$

$$\text{Maximum assets selected:} \quad \sum_{i=1}^n z_i \leq 30$$

Geographical allocation:

$$\text{US:} \quad 0.35 \leq \sum_{i \in \text{US}} w_i \leq 0.45$$

$$\text{Europe:} \quad 0.25 \leq \sum_{i \in \text{EU}} w_i \leq 0.35$$

$$\text{Asia/Emerging markets:} \quad 0.25 \leq \sum_{i \in \text{Asia/EM}} w_i \leq 0.35$$

Turnover Cost Constraints:

$$|w_i - w_{0,i}| \leq \Delta_i; \quad \forall i \in 1, \dots, n$$

$$\Delta_i \leq M \cdot z_i; \quad M = 1.0; \quad \forall i \in 1, \dots, n$$

$$\text{Total Cost Constraint:} \quad C_{\text{base}} \sum_{i=1}^n z_i + C_{\text{var}} \sum_{i=1}^n \Delta_i \leq \text{Cost}_{\text{max}}$$

Where:

- w_i is the individual weight of the i^{th} asset in our portfolio.
- $\gamma = \frac{1}{\lambda}$ is the risk-aversion parameter.
- Δ_i represents the absolute change in weight for asset i
- $z_i \in \{0, 1\}$: is a binary indicator variable (1 if asset i is traded, 0 otherwise).
- $\alpha_{\min}, \beta_{\max} = (0.01, 1.0)$
- Σ is the covariance matrix of asset returns.
- μ is the vector of expected asset returns.
- r_f is the risk-free rate of return. We took the US 1Y Treasury Bill = 3.62%

2 Explanation of the model

2.1 Past method

The main idea is to balance expected return against portfolio risk. Our most important formula is the Sharpe ratio, which outputs the excess return per unit of volatility:

$$\max_x \frac{\mu^T x - r_f}{\sqrt{x^T \Sigma x}}$$

For the intermediate report, we decided to change this non-convex problem into a convex quadratic one problem by using a change of variables :

$$y = \kappa x, \quad \kappa = \frac{1}{(\mu - r_f e)^T x}$$

By doing so, maximizing the original ratio becomes equivalent to minimizing:

$$y^T \Sigma y$$

This problem is called a Charnes-Cooper transform, and we get a convex quadratic programming problem, which can be solved efficiently.

What we didn't know at the time is that this model does not work when we input the turnover constraint with a fixed and variable components.

2.2 Actual method

The last model required the objective and constraints to be homogeneous. When we input fixed and variable turnover costs, the old model fails because our transaction costs are often non-linear and include fixed components. The transition to the current MIQP model was necessary to precisely account for these frictions.

By introducing binary indicator variables z_i , we can distinguish between a "no-trade" state and an "active-trade" state, allowing for the application of a fixed base cost C_{base} regardless of trade size, alongside a variable cost C_{var} proportional to the change in weight Δ_i . This degree of precision is required to ensure that the total turnover cost, calculated as $C_{base} \sum z_i + C_{var} \sum \Delta_i$, remains within the specified budget.

We implement a Mixed-Integer Quadratic Programming (MIQP) model that uses a grid search to find the portfolio that maximizes the Sharpe Ratio.

2.3 Equivalence between the Markowitz minimization and our maximization

A standard Markowitz problem minimizes risk σ_p^2 subject to a required return $\mu_p \geq R$. Using Lagrange multipliers, the return constraint is moved into the objective function, creating a penalized objective: $\min w^T \Sigma w - \lambda(\mu^T w - R)$.

By rearranging the terms and redefining the constant $\gamma = 1/\lambda$, the minimization of risk

minus return is equivalent to the maximization of return minus risk:

$$\max \mu^T w - \gamma w^T \Sigma w$$

Here, γ represents the penalty for variance. The logic being that for every point on the efficient frontier, there exists a specific γ that yields the same optimal weight vector w as a risk-minimization problem with a specific return target R .

In fact, the only difference in the maximization or minimization is the psychology of the person computing it. Minimization is a cost function and focuses on reducing risk as much as possible, while the maximization searches for the most return investors can expect with the amount of risk they're willing to take.

$$W^* = \arg \max \{\mu^T w - \gamma w^T \Sigma w\} = \arg \min \{\gamma w^T \Sigma w - \mu^T w\}$$

2.4 Convexity of our model

Even though the inclusion of binary variables ($z \in \{0, 1\}$) makes the feasible set non-convex, the Quadratic Term ($w^T \Sigma w$) remains Positive Semi-Definite (PSD).

Because Σ is PSD, the risk term is convex. Subtracting a convex term from a linear term ($\mu^T w$) results in a concave objective function, which is globally optimizable via MIQP solvers like CPLEX or Gurobi.

2.5 Post-Optimization Sharpe Ratio Calculation

In the final step of the process, the model evaluates the Sharpe Ratio for every candidate portfolio generated during the grid search.

The model iterates through the results of the utility maximization, calculates the Sharpe for each optimal weight vector and identifies the specific portfolio that yields the highest value. This two-step logic allows us to find the portfolio with the highest excess return per unit of risk while respecting the geographical, beta, and turnover cost constraints. This would be mathematically impossible in a single step.

3 Justification for choosing the Markowitz model

We chose to use the Markowitz model again for our portfolio optimisation mainly because of its conceptual simplicity (the trade-off between return and risk is easy to understand and apply) and its ability to be extended by adding constraints tailored to our project, as it is possible to add numerous practical constraints.

3.1 Additional constraints

We added the following constraints: geographical, maximum individual weight, market cap size, alpha and beta constraints. These adjustments allow us to obtain a more realistic and diversified portfolio, which is what we are aiming for.

3.2 Sharpe efficient portfolio

With our Markowitz model, we decided to take the maximum Sharpe efficient portfolio because we want to optimize the risk-return ratio over the five weeks of our portfolio (and because we want to win the student challenge). This efficient Sharpe portfolio seems more realistic and optimal to us because it incorporates both asset volatility (i.e. risk) and a higher return than the minimum variance portfolio, which only reduces total risk. It is therefore a portfolio located on the Markowitz efficient frontier that limits risk while optimising expected returns.

4 Resolution procedure

1. Parameter estimation

Every week, share price data is updated from the Google Sheet file. We then recalculate the expected return of each asset (average of daily returns over the last four weeks) and the covariance matrix of returns. We use Python to run the code because of its simplicity.

2. Model formulation

We use Markowitz's model with constraints. The objective is to determine the weighting vector $w = (w_1, w_2, \dots, w_n)$ that maximizes the Sharpe ratio.

Under the following constraints:

- Sum of weights = 1
- No short position
- Limit per asset
- Sectoral or geographical limit
- Maximum number of assets in the portfolio

These constraints ensure that the portfolio is fully invested, well diversified, long-only, and limited in its concentration on any single asset, sector, or geographical area.

3. Transaction cost management

From the second week onwards, we incorporate rebalancing costs into our optimisation to avoid excessive portfolio adjustments. This means that each week, a free budget of €100 is included and any overspending will be deducted from the value of the portfolio. This constraint encourages prudent management and limits unnecessary asset rotation.

4. Justification

This procedure is suitable because it is based on a robust and transparent model, can be easily adapted to automatic weekly updates, incorporates practical constraints, and can be easily represented in Python code.

5 Technical explanation of the portfolio optimization process with Python

5.1 Preparation of the data on Excel

We use excel to update fo each available asset, including our benchmark (MSCI world index), their average return, alpha and beta. Then, the covariance matrix is created using the data analysis function.

5.2 Data importing on Python

We then import all the data on Python using the pandas package. Constants such as the geographical allocation of each asset are hard coded as won't change weekly.

The covariance matrix is loaded and goes through extra verifications to ensure it is actually positive semi-definite.

5.3 Decision Variables and Turnover Logic

The model defines the portfolio weights as the primary decision variables. To manage turnover, it introduces binary variables (z) and continuous change variables (Δ). This allows the code to mathematically link weight changes to a specific cost structure: a fixed base cost for initiating any trade and a variable cost proportional to the trade size.

5.4 Constraint Enforcement

The solver enforces several strict limits: weights must sum to 100%, no single asset can exceed 10% of the portfolio (M_{max}), the total portfolio beta must remain below 1, and geographical balance must be respected.

5.5 The Two-Step Optimization Process

The code uses a grid search to find the maximum Sharpe Ratio. In the first step, it solves the MIQP maximization problem ($\max \mu^T w - \gamma w^T \Sigma w$) for 20 different risk-aversion levels (γ). In the second step, it calculates the Sharpe Ratio for each resulting portfolio and selects the one that offers the best risk-adjusted return.

5.6 Final Reporting and Export

Once the optimal portfolio is identified, the code validates that all turnover and allocation constraints are met. It generates a detailed report of the final metrics (expected return, alpha, beta, risk, and total transaction costs and exports the results to an Excel file.

6 Choice of data

As part of this project, all data used covers the period from September 2022 to the most recent date available, i.e. approximately three years of history. There are several reasons for this choice: economic, statistical and practical. In line with the project's objectives and the constraints of the Markowitz model.

Firstly, the use of recent data ensures that the estimates best reflect current market conditions. As the competition spans a short period of five weeks, the use of historical data that is too old

could become obsolete (macroeconomic shocks, changes in monetary policy, corporate restructuring). By focusing on a recent period, the model more accurately reflects the current financial situation of companies and thus improves the economic relevance of the portfolio.

Secondly, this period ensures that the data for all assets considered is complete. The starting point set at September 2022 ensures that each asset has a sufficient and consistent history, thus avoiding any selection or estimation bias.

Thirdly, this choice of data over the designated period aims to ensure statistical balance within the framework of a mean-variance model. Too short a period would make estimates of returns and, above all, covariances excessively noisy and unstable, while too long a period would dilute recent information relevant to weekly reallocation decisions. A three-year window is therefore a reasonable compromise, widely used in portfolio management practice, allowing for sufficiently robust estimates while remaining sensitive and consistent with current market conditions.

Finally, this choice is consistent with the dynamic structure of the project, which is based on weekly portfolio reoptimisation. The parameters (expected returns, volatilities and correlations) are recalculated regularly from this database, allowing the model to gradually adapt to market developments while limiting instability and excessive turnover in allocations. Thus, the chosen period directly contributes to the reliability, stability and economic performance of the optimised portfolio.

7 Choice of constraints

Since we must invest all our capital at an initial date rather than spreading our investments over time to reduce risk, we will apply strict but logical constraints.

7.1 Choice of markets

Our first constraint will be geographical. In theory, and according to the two articles by SIFMA and Visual Capitalist, the US accounts for around 50% of global market capitalisation, Asia for about 28%, Europe (including the UK) for roughly 15%, and the rest of the world (Africa, Middle East, etc.) for 7%.

We have decided to adjust these theoretical weights in order to outperform the market and to be less dependent on the US dollar, which we expect to depreciate against the euro.

Therefore, our first constraint will require the portfolio to consist of 40% US assets, 30% Asian (or emerging market) assets, and 30% European assets. This means increasing our exposure to Europe compared to the theoretical allocation, as we believe that the current economic environment offers genuine opportunities in Europe (such as lower valuations and fiscal stimulus policies).

To ensure that the model isn't too restricted, and to minimize the possibility of having to drastically change our allocations the following week, which would be expensive, we give the model a 5% room to maneuver.

7.2 Choice of Market Capitalisation size

We initially imposed a market-capitalisation constraint to focus primarily on mid- and large-cap companies, defined as those with market capitalisation above USD 2 billion.

In practice, this constraint proved difficult to implement. Obtaining consistent and reliable market-capitalisation data was challenging: the Yahoo API used in Excel did not cover all companies and presented inconsistencies across currencies. In addition, the constraint did not materially improve the model's accuracy and, in fact, reduced its flexibility. Consequently, we decided to remove this constraint entirely.

7.3 Choice of Benchmark

We chose the MSCI World index as our benchmark because it represents our portfolio quite well. The MSCI World Index captures large and mid-cap representation across 23 Developed Markets countries.

7.4 Choice of Beta

We will use the model with a market factor:

$$R_i = \alpha_i + \beta_i R_m + \epsilon_i$$

where :

- α_i is the regression intercept for action i ;
- ϵ_i are the residuals of action i .

This model is often referred to as Sharpe's Single-Index Model.

In our game, we will have to re-optimize a portfolio of 30 stocks every week. However, the covariances between stocks are difficult to estimate in a stable manner on a short weekly sample (last 3 years). To limit this instability issue, we adopt a factorial model with a single market factor, i.e. where each stock return is explained by its sensitivity to the market (β).

This model reduces estimation noise, allows for the reconstruction of a consistent covariance matrix each week in a much more simple manner, and naturally links equity risk to a relevant economic factor: the MSCI World Index.

We therefore impose a constraint on the total beta of the portfolio:

$$\beta_p = \sum_{i=1}^N w_i \beta_i \leq 1$$

With a $\beta \leq 1$ for our entire portfolio, by taking each asset weight multiplied by the β of each asset, we reduce our exposure to risk. Our portfolio is therefore more defensive than the overall market in the short term, which is consistent with our idea of reducing risk while seeking optimal returns.

7.5 Maximum individual weight

Let's add another metric, and not the least important one: the total weight of each asset in our portfolio. This is essential to diversify our portfolio and to ensure that the results of the Markowitz model are consistent. We followed the approach presented in the Optimisation in Finance syllabus:

To determine this limit m , we applied the following formula:

$$m = \frac{k}{N}$$

where N is the number of assets in the portfolio (30), so $1/N$ represents the average weight if all assets were equally weighted and k is a diversification factor. We arbitrarily chose $k = 3$ based on our research and the findings of DeMiguel et al. (2009).

This constraint therefore ensures that no single asset accounts for more than 10% of the portfolio ($w_i \leq 0.10$). It prevents the portfolio from being overly dependent on any one asset and promotes better diversification.

7.6 Positive Alpha Constraint

We would like to select a well-performing portfolio with positive momentum. We have chosen Alpha as a suitable measure of performance, as it compares the returns of a stock to a benchmark. The higher the alpha, the better the return of the asset. It means that it has outperformed the benchmark in the given time period.

We have decided to reduce the amount of historical data we take into account, for it is well known that positive momentum cannot continue eternally. In the long-term, alpha will go down. Therefore, we have shortened our horizon to 1 year for this metric.

We will add a constraint to ensure that the weighted sum of alphas in the portfolio is positive. We select 0.01 as a suitable threshold. This means that the portfolio has beaten the benchmark by at least 1% in the last year.

We compute the constraint like this:

$$\sum_{i=1}^N w_i \alpha_i \geq 0.01$$

7.7 Non-selection of a sectoral constraint

In order to avoid too many constraints that could undermine our Markowitz model, we decided not to implement a sector constraint for two reasons:

1. Firstly, in our opinion, sectors are cyclical over periods longer than one year, so they are of little interest for our five-week portfolio. Integrating a sector constraint would have required us to make several adjustments to this constraint during the year in order to track the correct evolution of these sectors.
2. Secondly, in the Markowitz model, the risk of each asset is measured by its correlation with other assets via the covariance matrix. Assets in the same sector have a strong positive correlation. The model, which aims to minimise the total variance of the portfolio, will automatically reduce the weighting of assets with a high correlation. This is why adding a sector constraint would have been redundant and would have reduced the efficiency and optimisation of our model in order to find our efficient portfolio.

7.8 Choice of Turnover

Imposing a strict turnover constraint in a Markowitz portfolio optimization framework is justified by both practical and theoretical considerations, Cornuejols and Tütüncü, 2006. This constraint is particularly important in large mean–variance optimization models, where the covariance matrix is often close to singular.

Practically, singularity can occur when:

- The number of assets is large relative to the length of the historical return series.
- Assets exhibit very high correlations.
- The data contains redundancy or estimation noise.

In such cases, small variations in input data can lead to disproportionately large changes in the optimal portfolio. By enforcing a strict turnover limit, we reduce this sensitivity, avoid excessive rebalancing during portfolio reoptimization, and improve the stability and practical implementability of the model's outcomes.

Instead of a turnover constraint, we implement transaction costs directly into the model Cornuejols and Tütüncü, 2006. Its advantages are : reduced sensitivity to new data and a more realistic management of our portfolio. In real life, any stock transaction (bid or ask), comes with costs. We are replicating a real constraint faced by portfolio managers.

8 Portfolio Analysis before the game

8.1 Geographical Allocation:

The portfolio is well-diversified geographically:

- United States: 37.5%

- Europe: 30%
- Asia / Emerging Markets: 32.5%

This balance between developed and emerging markets helps reduce region-specific risk.

8.2 Market Capitalization Allocation:

- **Large Cap & Mid Cap** : 90%
- **Small Cap** : 10%

The portfolio is mainly composed of large and mid cap stocks, providing stability. Small-cap exposure adds potential for higher growth.

8.3 Diversification & Constraints Compliance:

- Total assets selected: 18 (target: 30)
- Maximum individual weight: 10% (constraint respected)
- Weighted alpha: 0.2741 (well above the minimum of 0.01)
- Portfolio beta: 0.6553 (below the maximum of 1, implying lower volatility than the market)

All constraints are respected, resulting in a moderately risky portfolio with strong risk-adjusted return potential.

8.4 Expected Return & Risk:

- Expected return: 34.85%
- Volatility (Std Dev): 11.80%
- Sharpe ratio: 2.65

8.5 Key Observations:

- GE, 0941.HK, F9D.SI, and WMT each account for 10% of the portfolio, highlighting concentration on major regional and sector leaders.
- The portfolio combines stability (large-cap, low beta) with growth potential (small-cap, positive alpha).
- Geographical exposure is well-balanced, reducing dependency on a single market.

8.6 Portfolio Summary Table:

Ticker	Full Name	Region	Alpha	Beta	Weight (%)
GE	General Electric	US	0.47	1.32	10.00
0941.HK	China Mobile	Asia	0.17	0.17	10.00
F9D.SI	Fraser's Logistics & Ind. REIT	Asia	0.17	0.22	10.00
WMT	Walmart	US	0.19	0.61	10.00
PME.AX	Pinduoduo	Asia	0.38	0.94	8.15
BBVA	Banco Bilbao Vizcaya Argentaria	Europe	0.38	1.04	6.93
ABBV	AbbVie	US	0.09	0.20	6.70
ALIZF	Allianz	Europe	0.22	0.48	6.49
UCB.BR	UCB	Europe	0.27	0.47	5.64
NFLX	Netflix	US	0.31	1.28	5.46
BA.L	BAE Systems	Europe	0.16	0.20	5.09
ENGI.PA	Engie	Europe	0.12	0.20	4.73
0857.HK	AIA Group	Asia	0.20	0.18	3.36
PLTR	Palantir Technologies	US	0.50	2.14	2.29
SPOT	Spotify	US	0.35	1.38	1.69

Table 1: Portfolio composition with alpha, beta, and weight.

9 Evolution of the methodology during the competition

9.1 Choice of markets

During the game, our geographical allocation constraint was crucial to ensure stability in the outputs of the model.

Our geographical allocation moved as such :

- For European companies, our portfolio was exposed between 23% and 28%,
- for companies in the United States, allocations range between 39% to 43%,
- and for Asian/Emerging Markets companies, the share of the portfolio ranges between 31% and 33%.

Overall, no sudden move was noticed, emphasizing on the constraint's efficiency to minimize shocks.

9.2 Market Capitalization Allocation

As explained in section "Choice of Market Capitalization size", we did not move forward with this constraint.

9.3 Diversification & Constraints Compliance

All constraints cited in subsection 7.3 have been respected for the entirety of the game.

9.4 Expected Return & Risk

During the competition period, our initial approach to turnover control was revised to fully comply with the transaction cost rules defined by the professor. In the intermediate report, turnover was only constrained indirectly through a bound on weekly portfolio weight variations. While this ensured portfolio stability, it did not explicitly model transaction costs or their impact on portfolio performance.

Implementation of Explicit Transaction Costs

To address this limitation, we implemented an explicit transaction cost structure aligned with the competition rules:

- **Fixed cost:** 10 EUR per asset traded, regardless of transaction size
- **Variable cost:** 0.05% of the transaction amount, applied to both purchases and sales
- **Weekly budget:** 100 EUR free rebalancing allowance per week

The total transaction cost associated with a portfolio rebalancing is computed as:

$$\text{Cost}_t = 10 \sum_{i=1}^n z_i + 0.05\% \times V_t \sum_{i=1}^n \Delta_i$$

where $\sum_{i=1}^n z_i$ denotes the number of assets whose positions changed between two periods, Δ_i the difference in portfolio weight of asset i between t and $t - 1$, and V_t the total portfolio value at time t .

To find the total cost for the entire portfolio of n stocks, we sum the individual costs:

$$\text{Total Cost} = \sum_{i=1}^n C_i$$

Transaction costs exceeding the weekly free rebalancing budget of 100 EUR are directly deducted from the portfolio value, creating a direct penalty for excessive turnover.

9.5 Adjustment of Turnover and Transaction Cost Modelling

This transaction cost modelling led to a structural change in our optimization framework.

- **Before:** Turnover was constrained *ex ante* through a hard bound on total weight variation between two consecutive periods.
- **After:** Turnover arises endogenously from the optimization process, as the solver trades off expected portfolio improvement against transaction costs.

This formulation better reflects realistic trading conditions and naturally discourages unnecessary portfolio adjustments without relying on arbitrary turnover limits.

Implementation Challenges

The introduction of transaction costs, in particular the fixed cost component, increased the computational complexity of the optimization problem. The model effectively became a Mixed-Integer Quadratic Program (MIQP), which is more demanding to solve than a standard Quadratic Program (QP).

In some weeks, the solver was unable to identify a feasible solution satisfying both the optimization objective and the transaction cost budget constraint within reasonable computation time. This typically occurred when market conditions required substantial portfolio rebalancing or when several constraints were simultaneously binding.

In such cases, manual adjustments were applied to ensure feasibility and portfolio continuity. These included temporarily allowing transaction costs above the weekly budget, maintaining the previous portfolio when rebalancing costs exceeded expected benefits, or prioritizing only the most impactful trades. This pragmatic approach ensured continuous portfolio management throughout the competition period while preserving the core principles of cost-aware optimization.

9.6 Analysis Portfolio after the game

Ticker	Full Name	Region	Alpha	Beta	Weight (%)
UCB.BR	UCB SA	Europe	0.22	0.33	4.71
PLTR	Palantir Technologies	US	2.32	1.24	2.37
BBVA	Banco Bilbao Vizcaya Argentaria	Europe	0.96	0.53	8.58
ENGI.PA	Engie	Europe	-0.13	0.24	6.84
NFLX	Netflix	US	1.22	-0.41	2.38
GE	General Electric	US	1.60	0.50	10.10
WMT	Walmart	US	0.76	0.20	10.10
F9D.SI	Frasers Logistics & Commercial Trust	Asia	0.24	0.24	5.55
BA.L	BAE Systems	Europe	0.36	0.21	1.24
ABBV	AbbVie	US	0.53	0.10	7.88
ALIZF	Allianz	Europe	0.27	0.34	5.09
ACGBY	Agricultural Bank of China	Asia	0.60	0.23	0.00
0857.HK	AIA Group	Asia	-0.11	0.40	7.13
PME.AX	Pinduoduo	Asia	1.11	0.44	7.30
0941.HK	China Mobile	Asia	-0.05	0.22	10.10
NVDA	NVIDIA	US	1.99	0.76	3.52
IBE.MC	Iberdrola	Europe	-0.06	0.18	2.86
000333.SZ	Midea Group	Asia	0.15	0.13	2.74
IBM	International Business Machines	US	0.86	0.19	1.52

Table 2: Portfolio composition with weights at the final rebalancing date.

10 Results over the 5-week competition

Stock selected	ENTRY DATES FOR EACH ASSET	TERMINAL WEEK DATES FOR EACH ASSET	Return of each stock	Initial Weight (W1)	Final Weight (W5)	Delta Weight	Portfolio return
UCB.BR (W1-5)	222,8	241,2	8,26%	5,64%	4,71%	-0,93%	
PLTR (W1-5)	172,14	193,38	12,34%	3,29%	2,37%	-0,92%	
BBVA (W1-5)	21,61	23,03	6,57%	6,93%	8,58%	1,65%	
ENGL.PA (W1-5)	21,93	22,28	1,60%	4,73%	6,84%	2,10%	
NFLX (W1-5)	111,22	93,25	-16,16%	5,46%	2,38%	-3,08%	
GE (W1-5)	304,17	307,21	1,00%	10%	10,10%	0,10%	
WMT (W1-5)	102,54	112	9,23%	10%	10,10%	0,10%	
F9D.SI (W1-5)	1,75	1,77	1,14%	10,00%	5,55%	-4,45%	
BA.L (W1-5)	1806,5	1724	-4,57%	5,09%	1,24%	-3,85%	
ABVV (W1-5)	232,29	226,82	-2,35%	7,46%	7,88%	0,42%	
ALIZF (W1-5)	430	447,85	4,15%	6,49%	5,09%	-1,40%	
ACGBY (W1-2)	19,89	18,61	-6,44%	1,69%	0,00%	-1,69%	
0857.HK (W1-5)	8,96	8,09	-9,71%	5,08%	7,13%	2,05%	
PME.AX (W1-5)	248,83	217,37	-12,64%	8,15%	7,30%	-0,85%	
0941.HK (W1-5)	87,85	83,9	-4,50%	10,00%	10,10%	0,10%	
NVDA (W2-5)	186,86	180,99	-3,14%	0,00%	3,52%	3,52%	
IBE.MC (W2-5)	17,98	18,21	1,28%	0,00%	2,86%	2,86%	
000333.SZ (W5)	79,6	79,45	-0,19%	0,00%	2,74%	2,74%	
IBM (W2-5)	304,86	300,98	-1,27%	0,00%	1,52%	1,52%	
Total Portfolio Return							0,23%
						Portfolio Gains :	2 319,85 €
						Turnover Cost :	- 595,36 €
						Total Gains :	1 724,49 €
						Real Portfolio Returns :	0,17%

Figure 1: Table of Portfolio Returns

10.1 Interpretation of Portfolio Performance

Table 1 summarizes the performance and evolution of a diversified equity portfolio over the period from Week 1 (10/11/2025) to Week 5 (15/12/2025). It reports individual asset returns, portfolio weights at the beginning and end of the investment horizon, changes in asset weights, and each stock's contribution to the overall portfolio return.

Over the five-week horizon, the portfolio generated a modest positive return of 0.23%. After accounting for transaction costs, the net portfolio return declined to 0.17%. This result reflects a balance between gains from certain holdings and losses from others, highlighting the sensitivity of aggregate portfolio performance to both asset returns and portfolio weight allocation.

Several assets, including UCB, Walmart (WMT), BBVA, ENGIE, and PLTR, recorded positive returns and contributed favorably to overall portfolio performance. In contrast, stocks such as Netflix (NFLX), PME.AX, 0857.HK, and BA.L experienced notable negative returns, which exerted downward pressure on the portfolio. Assets combining relatively large portfolio weights with negative price movements had the most pronounced adverse effect.

Changes in portfolio weights were primarily driven by relative price movements rather than active rebalancing decisions. Securities with positive performance tended to increase their final weights, while underperforming assets saw their weights decline or were fully eliminated from the portfolio. This illustrates how portfolio composition evolves mechanically over time in response to asset-level performance.

Overall, the analysis underscores the joint importance of individual asset returns and portfolio weights in determining aggregate portfolio performance. Despite positive contributions from several holdings, the presence of multiple loss-making assets and transaction costs constrained the portfolio's final net return.

11 Lessons learned & strategic decisions for the future

11.1 Lessons learned

The five-week competition provided a realistic perspective on the gap between theoretical financial models and practical portfolio management. Several key challenges stood out:

- **Constraint Redundancy:** In the final week, we realized that the Large-Cap and Mid-Cap constraint had become redundant. The optimized portfolios naturally gravitated toward these assets, and removing the constraint simplified obviously the solver's task.
- **The Turnover Paradox:** We faced persistent issues with the turnover constraint. When strictly enforcing the €100 transaction cost limit, the Python solver often returned "Infeasible" or suggested to us a "zero-trade" strategy, which was not optimal. In a first time, we allowed the limit to be exceeded, which resulted in asset weights costing €180 or more sometimes. In the final week, we decided to completely relax the €100 transaction cost constraint to observe the behavior of the model. However, the resulting transaction cost did not change significantly, so we ultimately maintained our previous configuration.
- **Data Integrity and Cleaning:** We encountered technical issues due to data quality. Specifically, the asset **YZCFF** (=Yanzhou Coal Energy) should have been removed immediately manually but it was not done. Indeed, this asset had only zero returns and could have harmed our optimization since the model would like to pick him in the portfolio with his null covariance with the other assets. Furthermore, the presence of duplicate assets in the provided excel sheet added unnecessary noise to the optimization process. It would have been better to have clean data, even though we do not believe that this has caused us any harm in our case with our optimised portfolios.
- **The Evolution of Automation:** A significant turning point was the implementation of `market_data.py` in Week 4. Moving from manual data entry to an automated system that extracts the covariance matrix, expected returns (μ), alphas (α), betas (β), constraints, and other inputs from Excel files reduced human errors and saved considerable time. Instead of manually retrieving and pasting these elements into Python like we did in the first weeks, our semi-automated setup allowed the code to pull the required data directly from specific Excel cells. While calculations relied on existing Excel functions, we still needed to update the stock data weekly and to calculate the returns and the covariance matrix on it so the Python code could access the new results.

11.2 Strategic decisions for the future

In the event that we were asked to continue this optimisation in the future, to improve the robustness and performance of our optimization, we would implement the following strategies:

- **API:** After creating a semi-automatic data extraction, the next step would be to set up a system where the Python code can fully analyse Excel and extract all the necessary

information so that it can calculate all the important values itself without that we have to implement very specific functions in specific cells or some manual manipulations in Excel. This would create a direct API between Python and Excel.

- **Advanced Backtesting Framework:** We would develop a dedicated backtesting module to simulate our optimization over historical periods. This would help us determine if our final performance was due to the bad market conditions during the 5 weeks or due to a fundamental flaw in our parameter estimation.
- **Refining the Optimization Model:** Given that many groups struggled with Markowitz (6 groups were in negative vs 5 were in positive after the 5 weeks), in the future we might explore Black-Litterman to compare. We know that Markowitz is notoriously sensitive to small changes in input. Furthermore our code could be improved by replacing the hard turnover constraints with a penalty term in the objective function (e.g. transaction costs) to ensure a solution is always found.
- **Python Code Technical Enhancements:**
 - **Model-Code Consistency:** We could transition rather to a formal convex quadratic programming approach (such as the Charnes-Cooper transformation allows us) to improve efficiency and theoretical alignment even though it is more complex to understand and to compute.
 - **Cleaning Data Function:** We could implement a function in the code to automatically identify, merge, or remove duplicate assets and outliers like the YZCFF asset (had only zero returns) before the optimization begins.
 - **Linear Approximation Replacement:** We could replace binary indicators for turnover costs with a linear approximation to avoid complex MIQP calculations.

12 Use of Artificial Intelligence

We used ChatGPT to enhance our written English by reformulating sentences, correcting grammatical mistakes, and helping us articulate our ideas more clearly. We also used ChatGPT for coding and debugging purposes. He also helped us format our texts correctly for LaTeX.

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